NOVATICS · DETECTION · RASPBERRY PI 5

Pi 5 Person Detector

Real-time person detection on the Raspberry Pi 5 CPU. One small model, YOLOX-tiny, finds people in each camera frame; two video checks remove things that only look like people. No accelerator needed, and every part is free for commercial use.

~20 FPS live on the CPU | Person only | Apache 2.0 / MIT / BSD

HOW IT WORKS

From camera frame to "person"

1 camera

Newest frame

Pi HQ camera at 1280×720. A background thread keeps only the latest frame, so output never lags. Optional 180° turn for an upside-down mount

2 model

YOLOX-tiny int8

Input 416×256, the camera's 16:9 shape, so no grey padding. 29 ms per frame on 3 CPU cores

3 person

Person score ≥ 0.45

Only the COCO "person" class is read. 0.45 still catches people partly hidden behind objects

4 motion

Has it ever moved?

A box is accepted once something moved inside it, then followed while the person keeps still. Chairs and coats never move

5 confirm

3 of 5 frames

A person is reported after appearing in 3 of the last 5 frames, which removes one-frame flickers

Steps 4 and 5 apply to video. For a single photo, call detect(frame, video=False).

USE IT

In an application

```
from person_detector import PersonDetector

detector = PersonDetector()             # load once, one per camera
people = detector.detect(frame)         # frame: BGR image from OpenCV or the Pi camera
# [{"label": "person", "conf": 0.92, "box": [x1, y1, x2, y2]}, ...]
```

Options: `PersonDetector(conf=0.45, motion=True, confirm=True, threads=3)`. To read the camera too, use `camera.LatestFrame("auto", rotate=180)`. A web UI for testing (`app.py`) shows the live feed, the people in view and switches for each check.

MEASURED ON THIS PI · 2026-10-12

Speed

~20 FPS — live, camera + UI

Model 29–31 ms per 1280×720 frame

~10 FPS — det3's original file

Same weights at 416×416: 84 ms per frame

MEASURED ACCURACY

Does it find people, and only people?

TEST	RESULT
250 COCO photos of rooms with furniture, no people	Model alone: a "person" in 4 (a chair back, a wig, clothes, a figurine). Shown as a still scene with the motion check: **0**
250 COCO photos with people (IoU ≥ 0.5)	`74% of people ≥ 64 px tall found`
Person half hidden behind a box, Pi camera, 61 frames	**Found in every frame**, no false boxes
Nearly still person, Pi camera, motion check on	Accepted within ~1.5 s
416×256 model vs det3's 416×416 file, 63 images, 66 people	Same people found (0 missed, 0 extra), mean box IoU 0.93

CHOICES

Why it is built this way

DECISION	REASON
YOLOX-tiny, not YOLO11	Apache 2.0, free to ship in a product. Ultralytics YOLO11 is AGPL-3.0: a closed product would need a paid licence
416×256 input	Same detail as 416×416 for a 16:9 camera without the grey padding. About 3× faster on the CPU
CPU, no accelerator	Works on any Pi 5. The Hailo-8 is not needed for 20 FPS
Motion check, not a higher threshold	Look-alikes scored 0.66–0.70, as high as partly hidden real people. Movement separates them
No box merging	Merging head and legs boxes lost 65 real people in group photos and helped once. YOLOX never split a person on the camera

WHAT TO EXPECT

Known limits

A completely still person at start-up

DELAY

The motion check needs some movement. Someone already in view and motionless when the app starts is reported once they move slightly, usually within about 1.5 s.

If that matters: use `motion=False`, which accepts look-alikes again.

Moving cameras

SETUP

On a handheld, pan-tilt or drone camera everything moves, so the motion check stops filtering.

Fix: use `motion=False` there and rely on the 3-of-5-frame confirmation.

Small or distant people

ACCURACY

YOLOX-tiny is a small model. People under about 64 px tall in the frame are found less often.

Fix: place the camera closer, or train a larger model on your own footage.

Heat

SPEED

The model keeps 3 cores busy. Without cooling the Pi 5 slows itself down at about 85 °C.

Fix: fit the Active Cooler and check `vcgencmd get_throttled`.

RUN IT

Commands

```
cd ~/detection
python3 -m venv --system-site-packages .venv
.venv/bin/pip install -r requirements.txt
.venv/bin/python app.py --rotate 180        # web UI on http://<pi-ip>:8000
.venv/bin/python -m pytest -q               # tests
```

YOLOX-tiny (Megvii, Apache 2.0) · onnxruntime 1.31 (MIT) · OpenCV 5.0 (Apache 2.0) · Flask 3.1 (BSD) · Python 3.13 · Raspberry Pi 5, 16 GB